

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel International Advanced Level

Wednesday 08 May 2024

Morning (Time: 1 hour 30 minutes)

Paper reference **WCH11/01**

Chemistry

International Advanced Subsidiary/Advanced Level

UNIT 1: Structure, Bonding and Introduction to Organic Chemistry

You must have:
Scientific calculator, ruler

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- You will be assessed on your ability to organise and present information, ideas, descriptions and arguments clearly and logically, including your use of grammar, punctuation and spelling.
- A Periodic Table is printed on the back cover of this paper.

Advice

- Read each question carefully before you start to answer it.
- Show all your working in calculations and include units where appropriate.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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SECTION A

Answer ALL the questions in this section.

You should aim to spend no more than 20 minutes on this section.

For each question, select one answer from A to D and put a cross ☐. If you change your mind, put a line through the box ☒ and then mark your new answer with a cross ☐.

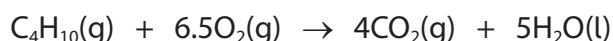
- 1 How many **atoms** are there in 10 g of methane?

[A_r values: H = 1.0 C = 12.0 $L = 6.02 \times 10^{23} \text{ mol}^{-1}$]

- ☐ A 3.76×10^{23}
☐ B 9.63×10^{23}
☐ C 1.88×10^{24}
☐ D 4.82×10^{24}

(Total for Question 1 = 1 mark)

- 2 What is the maximum volume of butane that can be completely burned in 39 cm^3 of oxygen at room temperature and pressure?



- ☐ A 6.0 cm^3
☐ B 12 cm^3
☐ C 24 cm^3
☐ D 45 cm^3

(Total for Question 2 = 1 mark)

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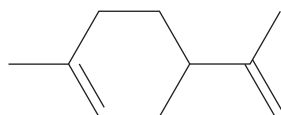
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3 This question is about limonene.



(a) What is the molecular formula of limonene?

(1)

- ☐ A C₁₀H₁₀
☐ B C₁₀H₁₆
☐ C C₁₀H₁₈
☐ D C₁₀H₂₂

(b) Limonene is completely hydrogenated by hydrogen in the presence of a nickel catalyst. What is the **empirical formula** of the product formed?

(1)

- ☐ A C₅H₁₁
☐ B C₅H₉
☐ C C₅H₈
☐ D CH₂

(Total for Question 3 = 2 marks)

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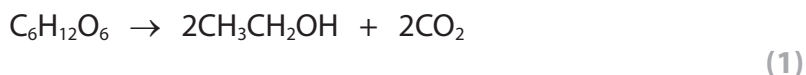
4 What is the best definition of a hydrocarbon?

- ☐ A a compound containing carbon, hydrogen and oxygen
- ☐ B a compound containing carbon and hydrogen only
- ☐ C a compound that contains single carbon to carbon bonds
- ☐ D a mixture containing carbon and hydrogen only

(Total for Question 4 = 1 mark)

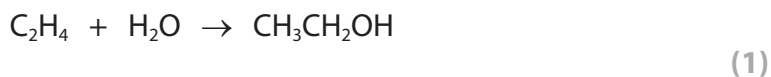
5 This question is about the production of ethanol by fermentation of glucose or by hydration of ethene.

(a) What is the percentage atom economy by mass for the production by fermentation?



- ☐ A 24.4%
- ☐ B 25.6%
- ☐ C 48.9%
- ☐ D 51.1%

(b) What is the percentage atom economy by mass for the production by hydration?



- ☐ A 39.1%
- ☐ B 60.9%
- ☐ C 78.2%
- ☐ D 100%

(Total for Question 5 = 2 marks)

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6 This question is about electronic configurations.

(a) What is the electronic configuration of an atom of calcium?

(1)

- ☐ **A** [Ne] $3s^1$
- ☐ **B** [Ar] $4s^1$
- ☐ **C** [Ne] $3s^2$
- ☐ **D** [Ar] $4s^2$

(b) What is the electronic configuration of a Cr^{3+} ion?

(1)

- ☐ **A** [Ar] $3d^24s^1$
- ☐ **B** [Ar] $3d^3$
- ☐ **C** [Ar] $3d^4$
- ☐ **D** [Ar] $3d^5$

(Total for Question 6 = 2 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.



7 This question is about bond angles.

(a) What is the expected bond angle in a boron trichloride molecule, BCl_3 ?

(1)

- ☐ **A** 90°
- ☐ **B** 107°
- ☐ **C** 109.5°
- ☐ **D** 120°

(b) What is the expected bond angle in an ammonium ion?

(1)

- ☐ **A** 90°
- ☐ **B** 104.5°
- ☐ **C** 107°
- ☐ **D** 109.5°

(c) What is the expected bond angle in an oxygen difluoride molecule, OF_2 ?

(1)

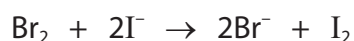
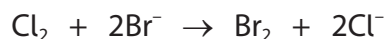
- ☐ **A** 90°
- ☐ **B** 104.5°
- ☐ **C** 109.5°
- ☐ **D** 180°

(Total for Question 7 = 3 marks)

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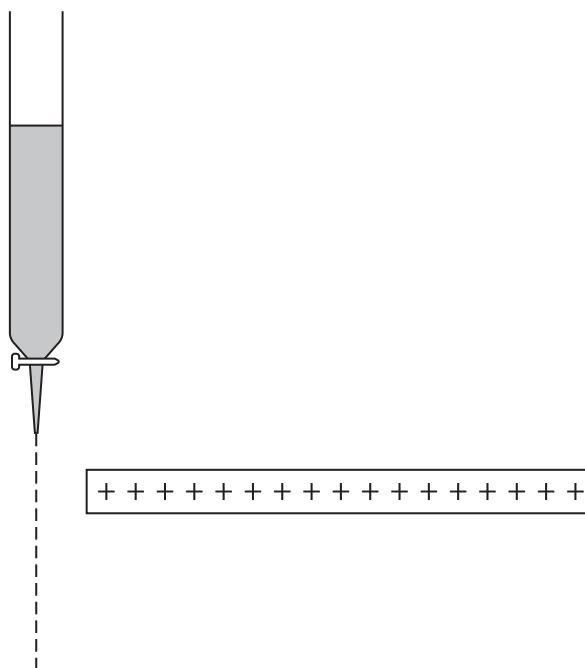
8 Which is the strongest **oxidising** agent in these displacement reactions?



- ☐ A chlorine
- ☐ B bromide ions
- ☐ C bromine
- ☐ D iodide ions

(Total for Question 8 = 1 mark)

9 To compare polar and non-polar liquids, a stream of liquid from a burette and a positively charged plastic rod can be used. Which row of the table shows the correct direction of deflection?



Direction of deflection		
	Polar liquid	Non-polar liquid
☐ A	towards the rod	away from the rod
☐ B	away from the rod	no deflection
☐ C	towards the rod	no deflection
☐ D	no deflection	towards the rod

(Total for Question 9 = 1 mark)

10 Which equation represents the **second** ionisation energy of magnesium?

- ☐ A $\text{Mg(g)} \rightarrow \text{Mg}^{2+}(\text{g}) + 2\text{e}^-$
- ☐ B $\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}^-$
- ☐ C $\text{Mg}^+(\text{s}) \rightarrow \text{Mg}^{2+}(\text{s}) + \text{e}^-$
- ☐ D $\text{Mg(g)} \rightarrow \text{Mg}^+(\text{g}) + \text{e}^-$

(Total for Question 10 = 1 mark)

11 This question is about the isoelectronic ions N^{3-} , O^{2-} , F^- , Na^+ , Mg^{2+} and Al^{3+} .

(a) Which ions are shown in order of **decreasing** ionic radius?

(1)

- ☐ A $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
- ☐ B $\text{Mg}^{2+} > \text{Na}^+ > \text{Al}^{3+}$
- ☐ C $\text{Al}^{3+} > \text{Na}^+ > \text{Mg}^{2+}$
- ☐ D $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$

(b) Which ions are shown in order of **increasing** ionic radius?

(1)

- ☐ A $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$
- ☐ B $\text{F}^- < \text{N}^{3-} < \text{O}^{2-}$
- ☐ C $\text{O}^{2-} < \text{N}^{3-} < \text{F}^-$
- ☐ D $\text{N}^{3-} < \text{O}^{2-} < \text{F}^-$

(Total for Question 11 = 2 marks)

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12 Which pair of compounds will **not** react to form a precipitate?

- ☐ A $\text{AgNO}_3(\text{aq})$ and $\text{CaCl}_2(\text{aq})$
- ☐ B $\text{Ca}(\text{OH})_2(\text{aq})$ and $\text{CO}_2(\text{g})$
- ☐ C $\text{Ba}(\text{OH})_2(\text{aq})$ and $\text{HCl}(\text{aq})$
- ☐ D $\text{KI}(\text{aq})$ and $\text{Pb}(\text{NO}_3)_2(\text{aq})$

(Total for Question 12 = 1 mark)

13 What is observed when iron(II) carbonate reacts with ethanoic acid?

- ☐ A colourless solution
- ☐ B colourless solution and effervescence
- ☐ C green solution and effervescence
- ☐ D green solution

(Total for Question 13 = 1 mark)

14 Which is the simplest ionic equation for an alkali reacting with an acid?

- ☐ A $2\text{H}^+(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l})$
- ☐ B $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{aq})$
- ☐ C $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
- ☐ D $2\text{H}^+(\text{aq}) + \text{O}^{2-}(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$

(Total for Question 14 = 1 mark)

TOTAL FOR SECTION A = 20 MARKS



SECTION B

Answer ALL the questions. Write your answers in the spaces provided.

15 This question is about silver.

- (a) Silver chloride decomposes in light. This reaction was used in the first photographic plates.

Write an equation for this decomposition.
Include state symbols.

(2)

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- (b) Silver can form in displacement reactions.

- (i) State **two** observations that are made when copper metal is added to silver nitrate solution.

(2)

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- (ii) Write an equation for this displacement reaction.
Include state symbols.

(2)



- (c) (i) Describe the bonding in pure silver metal.
Include a 2D diagram to show the arrangement of 12 of the silver particles involved.

(3)

- (ii) Pure silver cups are too soft so small amounts of copper are added to make an alloy.

Explain why copper makes the silver less malleable.

(2)

(Total for Question 15 = 11 marks)



16 This question is about silicon.

(a) (i) Define relative atomic mass.

(2)

(ii) Calculate the relative atomic mass of a sample of silicon, using the isotopic abundance data provided.
Give your answer to 3 significant figures.

Isotope	Abundance (%)
^{28}Si	91.07
^{29}Si	4.62
^{30}Si	3.00
^{32}Si	1.31

(2)



- (iii) In the mass spectrum of silicon, there is also a small peak at $m/z = 14$.
Deduce the formula of this particle.

(1)

- (b) Consider the elements Al, Si, P and S.

- (i) Explain the trend in the first ionisation energies of Al, Si and P.

(3)

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- (ii) Explain why sulfur does **not** follow this trend.

(2)

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(Total for Question 16 = 10 marks)

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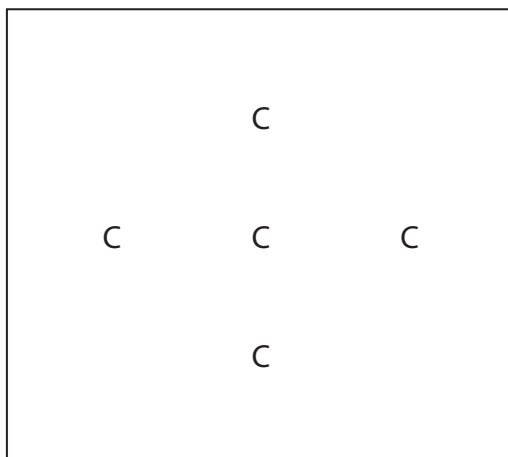
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17 Carbon exists as several different structures called allotropes.

- (a) Complete the dot-and-cross diagram by adding all the electrons in the outer shells of the five carbon atoms in a diamond tetrahedral unit.

(2)



- (b) Draw a labelled 3D structure of graphite, showing two layers of at least 13 carbon atoms each and the forces between the layers.

(2)

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- 18** Baking powder is added to cake mixtures to make cakes 'rise' by releasing a gas during cooking.

Baking powder contains an acidic derivative of tartaric acid and about 30% by mass of sodium hydrogencarbonate.

Baking powder releases the gas in two types of reaction during cooking, one of which is neutralisation.

- (a) (i) Give the name of the second type of reaction.

(1)

- (ii) Suggest why neutralisation does not occur in an unopened container of baking powder.

(1)

- (b) (i) The acidic derivative of tartaric acid contains 20.8% potassium, 25.5% carbon, 2.66% hydrogen by mass and the rest is oxygen.

Calculate the empirical formula.

(2)



- (ii) The acidic derivative of tartaric acid can be represented by the formula $\text{H}^+\text{K}^+\text{A}^{2-}$.

Write the equation for the neutralisation of sodium hydrogencarbonate by $\text{H}^+\text{K}^+\text{A}^{2-}$.

State symbols are not required.

(2)

- (iii) Calculate the maximum volume, in cm^3 , of carbon dioxide gas released by the neutralisation reaction in (b)(ii) from 5.00 g of baking powder, in an oven at 190°C at 101 000 Pa.

Baking powder contains 30% sodium hydrogencarbonate by mass.

$$[pV = nRT]$$

(5)

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- (iv) Calculate the decrease in volume when the carbon dioxide gas is cooled to 20°C.

(2)

(Total for Question 18 = 13 marks)

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19 Both butter and margarine are fats used in cooking.

(a) Margarine is sold as unsaturated fat.

State the meaning of the term unsaturated.

(1)

(b) (i) The degree of unsaturation can be determined by the reaction with bromine water.

Name the type of reaction.

(1)

Data using 0.5 g of some unsaturated fat in this bromination is given in the table.

Average number of C=C bonds per molecule	Volume of 0.0625 mol dm ⁻³ bromine water / cm ³
1	28.3
2	57.3
3	86.9
4	115

(ii) Plot a graph of the data. Include a line of best fit.

(3)



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(iii) Data for the bromination of 0.5 g samples of a margarine are shown.

Trial	Volume of $0.0625 \text{ mol dm}^{-3}$ bromine water / cm^3
1	36.9
2	34.1
3	39.3
4	32.5

Calculate the mean volume of bromine water using all results in the table.

(1)

(iv) Determine the average number of $\text{C}=\text{C}$ bonds per molecule of the unknown sample to 2 significant figures, using your graph.

(1)

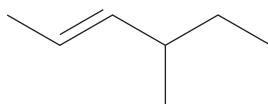


- (c) (i) Using the simplest alkene as an example, draw the mechanism to show the reaction with liquid bromine.

(4)

- (ii) Name this compound by applying IUPAC rules.

(1)



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- (d) Explain the meaning of "trans" in "trans" fats, using a **skeletal** formula showing four carbon atoms in your answer.

(2)

(Total for Question 19 = 14 marks)



20 Many synthetic polymers are used in the home.
Some are used as containers and others as coatings.

PVC and PTFE are two such examples.

(a) Draw the **displayed** formula of each of the monomers for these polymers.

(2)

chloroethene

tetrafluoroethene



- (b) Incineration and recycling are two methods of disposing of polymer waste.
Give **one** advantage and **one** disadvantage for each method.

(4)

Incineration

advantage

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disadvantage

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Recycling

advantage

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disadvantage

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(Total for Question 20 = 6 marks)

TOTAL FOR SECTION B = 60 MARKS
TOTAL FOR PAPER = 80 MARKS



28

* Lanthanide series

* Actinide series

Elements with atomic numbers 112-116 have been reported but not fully authenticated

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